

Clinical /Medical Decision Analysis (CDA)

Clinical /Medical Decision Analysis (CDA): Modelling test-treatment protocol/ decision to maximize patient's utility

To face uncertainties in diagnosis and treatment, clinical decision analysts developed strategies to assess the value of diagnostic technologies and treatment by combining decision theory and epidemiologic information. In many diagnostic situations the price a patient pays for misdiagnosis is much higher in two directions:

1. False Positive (FP), e.g., a FP diagnosis, untreatable target disorder or treatable in a patient who does not have it.
2. False Negative (FN) results, e.g., a clear bill of good health in a patient who does, in fact, has the target disease or disorder.

And sometimes the diagnostic test itself may involve some risk of death and/or discomfort or pain.

Decision analysis is a diagnostic strategy for coping with these rare but perplexing diagnostic problems.

The CDA is defined as a method of describing complex clinical problems in an explicit fashion, identifying the available courses of actions (both diagnostic and management), assessing the probability and utility(value) of all possible outcomes, and then making a simple calculation to select the optimal course of action.

Stephen Pauker and Jerome Kassirer (1981) identify six essential steps in a CDA:

Step 1: Problem formulation and creation of a decision tree.

Step 2: Assign probabilities to each branch that sprout from chance nodes.

Step 3: Assign utilities to each of the potential outcomes on the tree.

Step 4: Combine probabilities and utilities for each node on the decision tree.

Step 5: Then pick the decision that leads to the highest expected utility.

Step 6: Finally, test the decision for its sensitivity to clinically sensible change in probabilities and utilities.

Now, we introduce the six-step programme of CDA with an example.

Let us consider a patient who is 56 years old man with “heart-burn” for several years, may have coronary risk factors, and a 6 weeks history of non-exertional squeezing chest pain deep into lower sternum and epigastrium.

He has a negative physical examination and clinicians place esophageal spasm at the top of diagnostic hypothesis, judging his probability for coronary artery stenosis at 5%.

The sensitivity and specificity of the Exercise- ECG are 60% and 91% respectively and Table-1 below summarizes the results of such testing among patients in whom the prevalence or prior probability of significant coronary artery stenosis is 5%.

		>70% Coronary Stenosis			
		Present	Absent		
E-ECG	+ve	30	86	a+b	116
	-ve	20	864	c+d	884
		50	950	total= 1000	

Table 1: A 2*2 counting table depicting E-ECG results when prior probability of significant coronary stenosis is 5%

A test with a high true positive (TP) and low false negative (FN) role is said to be a sensitive test; a test with a low false positive (FP) and high true negative (TN) rate is said to be a specific test. A sensitive test is very good at detecting patients with the disease; a specific test is very good at screening out patients who do not have the disease. Thus, the test sensitivity applies to patients with the disease; test specificity applies to patients without the disease. Therefore, sensitivity and specificity are two separate measures.

$$\text{Sensitivity} = P [T+ | s] = 0.60$$

$$\text{Specificity} = P [T- | h] = 0.91$$

$$\text{FN} = 1 - \text{Sensitivity} = 1 - P [T+ | s] = P [T- | s] = 0.40$$

$$\text{FP} = 1 - \text{Specificity} = 1 - P [T- | h] = P [T+ | h] = 0.09$$

Now, we calculate unconditional result probabilities:

$$P [T+] = P [T+ | s] * P_s + P [T+ | h] * P_h = 0.6*(0.05) + 0.09*(0.95) \approx 0.12$$

$$P [T-] = P [T- | h] * P_h + P [T- | s] * P_s = 0.91*(0.95) + 0.4*(0.05) \approx 0.88$$

Now we calculate the posterior probabilities or the predictive values by using the Bayes Theorem:

$$PV^+ = P [s | T^+] = \frac{P [T^+ | s] * Ps}{P [T^+]} = \frac{0.6*(0.05)}{0.12} = 0.26$$

$$PV^- = P [h | T^-] = \frac{P [T^- | h] * Ph}{P [T^-]} = \frac{0.91*(0.95)}{0.88} = 0.98$$

And the corresponding complements are:

$$P [h | T^+] = 1-0.26 = 0.74$$

$$P [s | T^-] = 1-0.98 = 0.02$$

Using the values of the revised probabilities we implement the six steps of CDA that apply to this diagnostic problem.

Step 1: We create a “decision tree” or map all the pertinent courses of action and other consequences.

In the patient case we could send him for an E-ECG or withhold it. If we withhold it, we will fail to diagnose significant coronary stenosis 5% of the time but will be correct in discarding the diagnostic hypothesis 95% of the time. If we send him for the test, it can come back positive or negative. Referring to Table-1, if it comes back positive, it could be true positive (cell a) or a false positive (cell b). If E-ECG comes back negative, it could be a true negative (cell d) or a false negative (cell c). This course of actions and then consequences are shown in the decision tree in Figure-2 below:

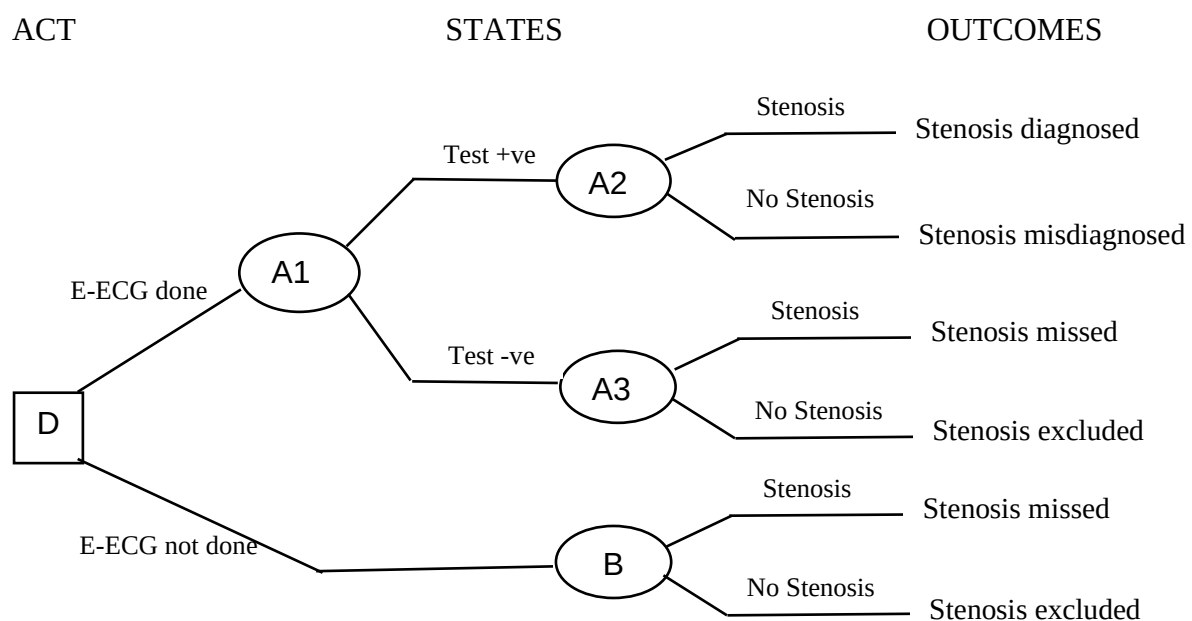
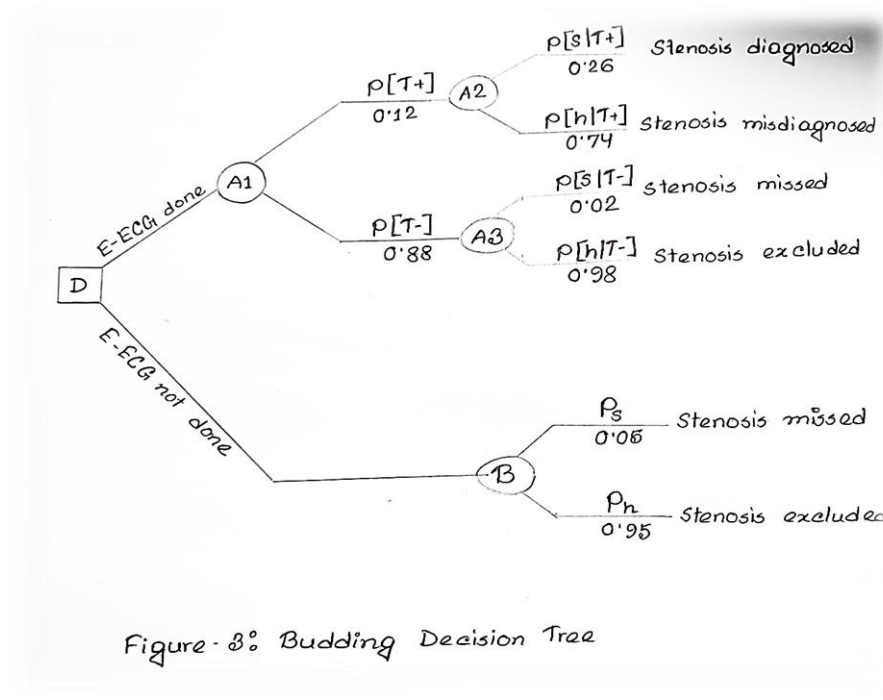


Figure 2: Decision Tree: This decision tree maps out all the options and outcomes we have just described above

Step 2: Assign Probabilities to the branches that sprout from each “chance node”



The branches from chance node B simply reflect the prior probability of significant stenosis ($P_s = 0.05$) and its complement. Those sprouting from A1 are proportion of such patients who will have positive $P[T+] = 0.12$, and negative $P[T-] = 0.88$ test results. The branches sprouting from chance node A2 are the proportion of those patients with positive E-ECG who do, $P[s|T+] = 0.26$ and who do not $P[h|T+] = 0.74$ have significant stenosis, and those sprouting from A3 are the proportion of patients with negative tests who do $P[s|T-] = 0.02$ and do not $P[h|T-] = 0.98$ have significant stenosis. All these probability values have been added to our decision tree and the results are shown in Figure-3.

Step 3: Assign utilities to each of the potential outcomes shown on the decision tree.

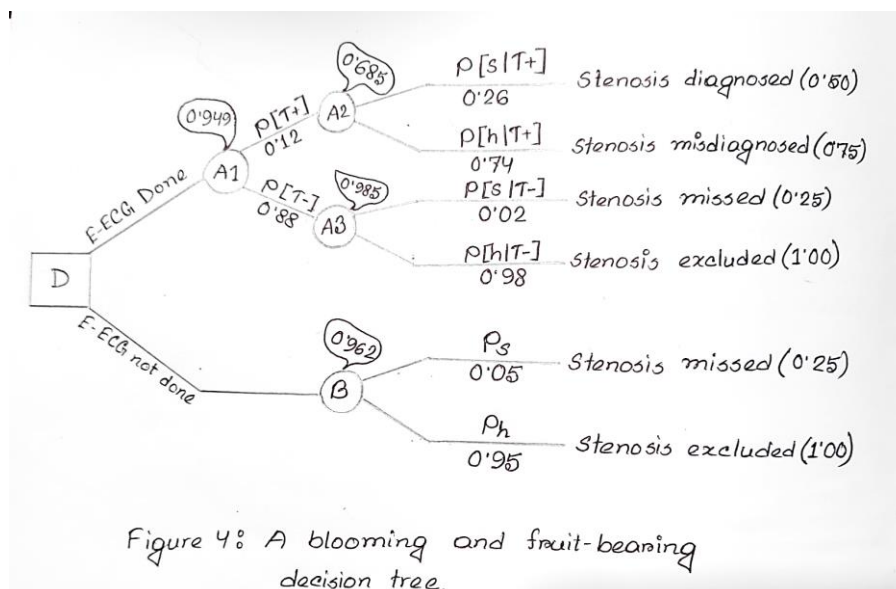
Clinicians, patients and even hospital administrators agree that some health outcomes are better than others. The QALY measurement corresponding to each outcome state is possibly the most accepted method of comparative judgement of various outcomes. All that decision analysis requires is for us to do our best to quantify those utility values of the different outcomes of the decision tree on a common interval scale (may be done by SG or TTO method).

We rank the outcomes from best to worst. In our example, we believe all will agree with the following ranking:

Best ↓ Worst	1	Stenosis excluded	=1.00
	2	Stenosis misdiagnosed	=0.75
	3	Stenosis diagnosed	=0.50
	4	Stenosis missed	=0.25

In ideal circumstances, we can go to the clinical literature and track down data on both survival duration and HRQoL for the outcomes at the end of the tree.

Step 4: Combine the probabilities and utilities/QALYs in each node on the decision tree.



Now all the averaging out and folding back has been added as balloons to our decision tree and the results are shown in figure-4.

Step 5: Pick the decision that leads to the highest (maximum) expected utility on QALY values.

In our example, the decision to send such patient for E-ECG leads to an EU=0.949, whereas the decision not to send him for E-ECG leads to an EU=0.962 and such patients are better off if the physicians do not subject them to E-ECG test. We have pruned the branch to show the exclusive of the choice.

Step 6: Sensitivity analysis, i.e., test our decision for its vulnerability to clinically sensible changes in probabilities and utilities.

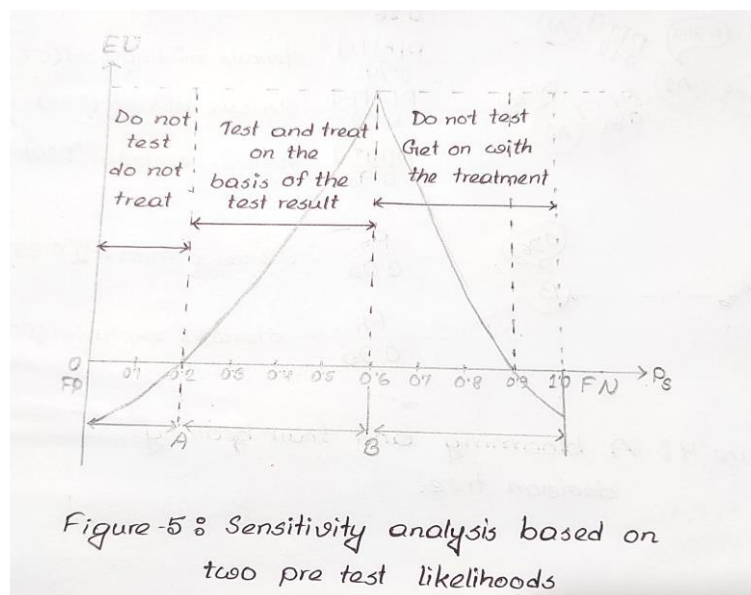
Would we still reject E-ECG if the prior probability of disease was 10% rather than 5%; if the utility for the outcome of stenosis misdiagnosed was 0.90 rather than 0.75? If we included a small (0.01%) but awful (utility or QALY=0.00) risk of dying during the E-ECG test?

No decision analysis is complete until we have tested it to see whether it resists credible changes in probabilities and utilities. If it does, the preferred course of action is clearly reinforced. If, on the other hand, credible changes in probabilities or utilities switch the decision, or if the EUs for the alternatives become so close that we are end up in a toss-up situation.

Clinicians with wide experience in applying the decision analysis have documented $\frac{1}{6}$ th of them wind-up as toss-ups. For our part, we are more impressed by the ability of decision analysis to handle $\frac{5}{6}$ th of tough decisions than by its failure to resolve the $\frac{1}{6}$ th that are unresolvable.

We shall close this discussion by intercepting one important extension which permits us to define specific prior probability of disease at which our diagnostic and treatment decision should switch. This is the “threshold” approach uses to assist the decision analysts about whether to carry out diagnostic tests (and guided by them) on the one hand, as whether to make therapeutic decisions without them.

This strategy suggested by Pauker and Kassirer, identifies two pre-test likelihoods and thereby these clinical decisions, as shown in Figure-5 below:



If the prior is less than cut off A, then we do not test and do not treat. Because in this region prior is so low that most +ve test results are likely to be false. But if prior is judged to be greater than A but less than B, then test and treat according to the test results. In this range of priors, test results will be mostly highly valued. If prior is greater than cut off B, we call it test-treatment threshold and we ought to assume that the patient has the target disorder and get on with the treatment.